

Project Development Phase Performance Test

Date	17 Feb 2026
Team ID	LTVIP2026TMIDS73343
Project Name	Heart Disease Analysis
Maximum Marks	5 Marks

Model Performance Testing:

S.No	Parameter	Screenshot / Values
1.	Data Rendered	I successfully connected my clinical dataset to MySQL and Tableau. I verified that all 4,500 records were correctly imported by running row-count queries in MySQL Workbench and checking the data pane in Tableau to ensure every column was visible and ready for analysis.
2.	Data Preprocessing	I performed data cleaning by running SQL DELETE commands to remove invalid records, such as patients with a BMI of 0 or negative sleep hours, which would have ruined the accuracy of my charts. In Tableau, I standardized the data by assigning proper data types (e.g., Number, String) and renaming columns to make them easier for a healthcare professional to understand.
3.	Utilization of Filters	I implemented interactive filters, such as Age Category, Sex, and Smoking Status, to make the dashboard functional for doctors. These filters allow a user to dynamically update all 10 visualizations at once to see risk trends for specific patient groups, like "Males aged 18-24".
4.	Calculation fields Used	I created three specific calculation fields in Tableau to add deeper clinical insights to my project: 1) BMI Categories: I transformed raw BMI numbers into easy-to-read groups like "Underweight," "Normal," and "Obese". Formula: <pre>IF [BMI] < 18.5 THEN "Underweight" ELSEIF [BMI] < 25 THEN "Healthy Weight" ELSEIF [BMI] < 30 THEN "Overweight" ELSE "Obese" END</pre>

		2) Risk Scores: I developed a custom score to categorize patients based on their heart disease probability. Formula: (IF [Smoking] = 'Yes' THEN 1 ELSE 0 END) + (IF [Stroke] = 'Yes' THEN 2 ELSE 0 END) + (IF [Diabetic] = 'Yes' THEN 1 ELSE 0 END) 3) Percentage Changes: I created this field to calculate the proportional difference in clinical metrics across different patient groups. Formula: (IF [Smoking] = 'Yes' THEN 1 ELSE 0 END) + (IF [Stroke] = 'Yes' THEN 2 ELSE 0 END) + (IF [Diabetic] = 'Yes' THEN 1 ELSE 0 END)
5.	Dashboard design	I built two comprehensive dashboards that integrate 10 distinct visualizations. These dashboards combine various chart types, such as bar charts, pie charts, and heatmaps, to provide a clear overview of clinical heart disease risks. The 10 visualizations included in my analysis are: 1. Gender Vs Heart Disease 2. Age Vs Heart Disease 3. Diabetic Vs Stroke 4. Impact of Smoking and Alcohol on Heart 5. Stroke Vs Other Disease 6. Race wise heart disease 7. General Health Vs Heart Disease 8. Physical Activity Vs Heart Disease 9. Age Vs BMI Vs Diabetic 10. People Got Stroke Suffering from Diabetes and Heart Disease
6	Story Design	I organized my analysis into a narrative flow using 10 individual visualizations. This allows a viewer to see the project's progress from basic demographic data to complex clinical health risks. The sequence of my analysis is as follows: 1. Gender Vs Heart Disease: Identifying risk based on gender. 2. Age Vs Heart Disease: Analyzing how age affects heart health.

		3. Diabetic Vs Stroke: Examining the link between diabetes and stroke. 4. Impact of Smoking and Alcohol on Heart: Evaluating lifestyle choices. 5. Stroke Vs Other Disease: Comparing stroke rates with other health conditions. 6. Race wise heart disease: Studying heart disease trends across different races. 7. General Health Vs Heart Disease: Seeing how personal health ratings correlate with disease. 8. Physical Activity Vs Heart Disease: Measuring the impact of exercise. 9. Age Vs BMI Vs Diabetic: A three-way analysis of age, body mass, and diabetes. 10. People Got Stroke Suffering from Diabetes and Heart Disease: A final look at high-risk patients with multiple conditions.
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